

From marketing and retail to healthcare and finance, adoption of artificial intelligence (AI) and machine learning (ML) in these sectors is drastically transforming the conventional approach for different applications. AI makes it possible for systems to sense, comprehend, act, and learn for performing complex tasks such as decision making that earlier required human intelligence. Unlike the regular programming, where action needs to be defined for every situation, AI in conjunction with ML algorithms can process large data sets, be trained to choose how to respond, and learn from every problem it encounters to produce more accurate results.

This has not only impacted how we use data but also how we design and fabricate hardware chips or integrated circuits (ICs) for enabling next-generation devices, thereby opening new opportunities. The growth of AI shifts the core of innovation from the software back to hardware. For better performance, for example, AI needs more memory as compared to traditional requirements to process and transfer large data sets. Consider the case of virtual assistants that are increasingly being used in homes. Without reliable hardware for such functionalities as those associated with memory and logic, these cannot work properly.

According to Accenture Semiconductor Technology Vision 2019, from the annual report by the Accenture Labs and Accenture Research, the semiconductor industry is highly optimistic about the potential of AI in their work in the coming years

when compared to that in technologies like extended reality, distributed ledgers, and quantum computing. Three-quarters (77 per cent) of semiconductor executives surveyed for the report claimed that they either had already adopted AI within their business or were piloting the technology.

The concept of AI and ML

Although the terms AI and ML are used interchangeably, the former aims at the success of a task, whereas the latter ensures accuracy. So, solving a complex problem is done through AI training, but maximising efficiency by learning from the data and already performed tasks is the concept of ML. ML relies on large datasets to find common patterns and makes predictions based on probability.

Applied AI is more commonly seen in systems that are designed for financial market predictions and autonomous vehicle control. Generalised AIs to handle different general situations is where ML, usually considered the subset of AI, comes into play. In what is called supervised learning ML algorithm, the relationship model between input data and target output is established to make predictions, whereas unsupervised learning does not involve categorising data for training. In situations like competing with humans in complex computer games where information needs to be gained from the environment continuously, reinforcement learning is implemented.

Sub-branches of ML like deep neural networks have already been

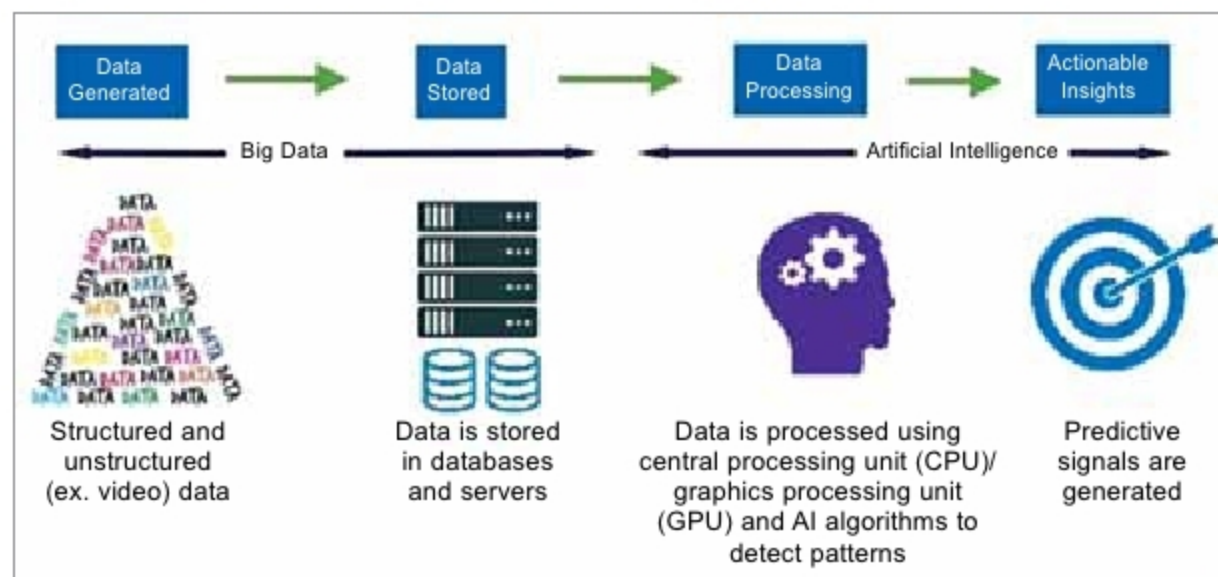
applied to fields, including speech recognition, social network filtering, computer vision, natural language processing, and so on. These technologies take and examine thousands of users' data for precision and accuracy applications like face recognition. This is contributing to the rapid development of innovations that are considered magical right now, but with hardware advancements, its place might be taken by much more advanced innovations in the coming decades.

How AI and ML applications are redefining the traditional systems

AI and ML could grow to the current extent due to advancements in not only algorithms but also in storage capabilities, computation capacity, networking, and the like, which made it possible to make advanced devices accessible to the masses at an economical cost. Traditionally, logic was usually hard-wired in the design of electronic systems. But in the light of high manufacturing costs presently and the growing complexity of chip development, AI-driven processor architectures are redefining traditional processor architectures to suit new demands.

Computation is mainly done on the central processing unit (CPU), the brain of the computer. With the emergence of computationally demanding applications that apply AI and ML algorithms, additional processing choices through combinations of graphics processing units (GPUs), microprocessors (MPUs), microcontrollers (MCUs), field-programmable gate arrays (FPGAs), and digital signal processors (DSPs) are coming up to meet the optimum feature requirements. The options, considered distinct categories, are gradually becoming heterogeneous processing solutions like system-on-chips (SoCs) and custom-designed application-specific integrated circuits (ASICs).

Unlike traditional storage methods to store parameters while tuning a neural network model during training and inference, high-bandwidth RAM is required not only at data centres but also for edge computing.



The process (Credit: <https://cooltechnospy.com>)